1 Chapter 2.1

Problem 4. *Consider the possible functions $f : [7] \to [9]$*

(a) How many have $f(3) = 8$?

(b) How many have $f(3) \neq 8$?

(c) How many have $f(1) \neq 5$ and are one-to-one?

(d) How many have $f(i)$ even, for all i?

(e) How many have rng$(f) = \{5, 6\}$?

(f) How many in which f^{-1} is not a function?

Answer:

(a) $f(3)$ has 1 choice. This leaves 6 other values that can go into 9 other bins. I have nine bins, that I can place six values in, with repitition allowed. This is 9^6.

(b) $f(3)$ has 8 choices, while the six other values have 9 choices. This is $8 \cdot 9^6$.

(c) $f(1)$ has 8 choices, $f(2)$ has eight choices (since one was removed by $f(1)$.) Then $f(3)$ has seven choices, $f(4)$ has six choices... This is $8 \cdot (8)_6 = \dfrac{8 \cdot 8!}{(8-6)!}$.

(d) Instead of 9 choices, we now have 4 choices. So 7 values have 4 choices. This is 4^7.

(e) Now we have only two choices, $\{5, 6\}$ for 7 values. This is 2^7. But, we must guarentee at least one values in $\{5\}$ and one value in $\{6\}$. So subtract out when all the values map to 5 or all the values map to 6, which is two cases. Final answer is $2^7 - 2$.

(f) For f^{-1} to be a function, the original function f must be one-to-one. We are looking for where the original function is not one-to-one. Base options is 9 choices for 7 values. So 9^7. But, we need to subtract out the one-to-one option, which is $(9)_7$. The final answer is $9^7 - (9)_7$.

Problem 7. *Find the number of onto functions $[k] \to [4]$.*

Answer: I am assuming that $k \geq 4$. The base case is 4^k options for distribution. Then need to subtract out the following:

(a) Missing 3 of the elements in [4]. Answer is $4^1 = 4$.

(b) Missing 2 of the elements in [4]. You have $\binom{4}{2} = 6$ elements to miss. You then have 2^k ways to distribute the values to the remaining elements. Each distribution has 2 ways that all the values end up on one elements instead of both, so remove those. The answer is $6(2^k - 2)$.

(c) Missing 1 of the elements in [4]. You have $\binom{4}{3} = 4$ different elements to miss. Once you miss, you can select 3^k places to go. But you need to guarentee that you don't miss more than 1 element. There are 3 ways to miss 2 elements with 2^k places to go. There are 3 ways to miss 3 elements with 1^k places to go. The answer is $4(3^k - 3(2^k - 2) - 3 \cdot 1^k)$

Final answer is $4^k - (4 + 6(2^k - 2) + 4(3^k - (3 + 3(2^k - 2))))$

Problem 11. *Give a combinatorial proof. For $n \geq 1$ and $k \geq 1$, then $2^{kn} > \max\{n^k, k^n\}$. Hint: Compare relations to functions.*

Answer: Before diving into the details with the general case, I'm going to do a numerical example to get us warmed up. Let us define two sets.

$$K = [4] = \{1, 2, 3, 4\} \quad N = [3] = \{1, 2, 3\}$$

All the possible relations from K to N are contained in the Cartesian product of $K \times N$.

$$K \times N = \{\{1, 1\}, \{1, 2\}, \{1, 3\} \ldots \{4, 1\}, \{4, 2\}, \{4, 3\}\} \tag{1}$$

Every possible relation on $K \to N$ is a subset of $K \times N$. For instance, for the relation $>$ would have $\{\{2, 1\}, \{3, 1\}, \{3, 2\}, \{4, 1\}, \{4, 2\}, \{4, 3\}\}$. So all the possible relations is $\mathscr{P}(K \times N)$, the power set of the cartesian product of $K \times N$. So, the number of possible relations is $2^{|K \times N|}$. If $k = |K|$ and $n = |N|$ then it is 2^{kn}.

Now, we focus our attention to functions instead of relations. The basic defintion of a function is that every input can only have one output. Thinking about this as distributions, the first number in K has 3 potential choices. The second number in K also has 3 choices, and so one. As a result, when looking at functions, there are $n^k = 3^4 = 81$ possible functions from $K \to N$. Likewise, if we go the opposite way, from $N \to K$, there are $k^n = 4^3 = 64$ possible functions.

What this is saying, is the number of possible relations between two sets will always be bigger than the possible functions from one set to another and vice a versa.

Proof. Suppose that $k, n \geq 1$ and K, N are sets of the first postive integers k, n denoted as $K = [k]$ and $N = [n]$. Then the set of all possible relations between K and N is the powerset of $K \times N$, denoted as $\mathscr{P}(K \times N)$. The cardinality of the powerset is $|\mathscr{P}(K \times N)| = 2^{kn}$, or the number of possible relations.

For a relation to be considered a function, every value in the domain must map to only one value in the codomain. The function $f : K \to N$ has n^k possible variations. The set of all function variations is a subset of all the relation variations. As such, $n^k \leq 2^{nk}$.

The inequality can be written $n^k < 2^{nk}$ by recognizing the empty set is inside the powerset but will never be considered to meet the definition of a function.

Now reversing the roles of K and N with the function $f : N \to K$ which has k^n possible variations. The same argument from above is used to prove $k^n < 2^{nk}$. This proves that $2^{kn} > \max\{n^k, k^n\}$. $\square$

2 Chapter 2.2

Problem 4. *Give combinatorial or bijective proofs of the following. Part of your job is to determine all values of n, k, and / orm for which the identities are valid.*

(a) $3^n = \sum_{k=0}^{n} \binom{n}{k} 2^{n-k}$

(b) $\binom{n}{k}\binom{k}{j} = \binom{n}{j}\binom{n-j}{k-j}$

(c) $\left(\!\binom{n}{k}\!\right) = \left(\!\binom{k+1}{n-1}\!\right)$

Problem 5. *What does* $\binom{n-1}{k-1} + \binom{n-2}{k-1} + \binom{n-3}{k-1} + \cdots + \binom{k-1}{k-1}$ *equal? Make a conjecture and then give a combinatorial proof.*

Problem 7. *Determine the number of solutions to each of the following equations. Assume all z_i are nonnegative integers unless stated otherwise.*

(a) $z_1 + z_2 + z_3 + z_4 = 1$

(b) $z_1 + z_2 + 10z_3 = 8$

(c) $z_1 + z_2 + \cdots + z_{20} = 401$ *where each* $z_i \geq 1$.

3 Chapter 2.3

Problem 2. *Rawr*

Problem 7. *Rawr*